DATA STRUCTURES & ALGORITHMS (C++)

Lab Problem Set: Searching & Sorting

The Scenario: Dhaka Anime Expo Ticket Management

You are volunteering as the lead programmer for a massive anime convention in Dhaka. To manage the entry queue cleanly, security has collected an unsorted list of unique **4-digit Ticket IDs** from attendees as they arrived at registration booths.

The system stores these IDs in a standard fixed-size dynamic C++ structure or raw array:

```
int tickets[] = {5420, 1102, 9981, 2301, 4455, 1101, 8832};
```

Assignment Tasks

Part 1: The VIP Alert (Linear Search)

An urgent announcement comes over the walkie-talkie: *"Ticket ID **2301** just won the mega raffle prize! Find out immediately if they are standing in our entry queue."*

Your Objective: Implement a sequence lookup routine using **Linear Search** logic to locate the position of Ticket ID `2301` inside the unsorted array.

- Provide the complete C++ function signature and implementation matching the condition.
- State the number of logical element comparisons executed before finding a valid match.

Part 2: Queue Structural Reorganization (Sorting)

The convention security gate requires the ticket stream arranged in strict **ascending order** (lowest ID to highest ID) before bulk check-in processing begins. You must analyze two separate structural sorting techniques:

Task 2A (Bubble Sort): Walk through the array elements using raw unoptimized Bubble Sort. Write the full C++ logic and fill out the trace for **Pass 1** in the table below.

Bubble Sort Pass 1 Trace Worksheet:

Step	Positions Compared	Elements Checked	Swap? (Y/N)	Resulting Array State
1	(0, 1)	5420 & 1102		
2	(1, 2)			
3	(2, 3)			
4	(3, 4)			
5	(4, 5)			
6	(5, 6)			

Task 2B (Selection Sort): Implement Selection Sort to arrange the data directly. Provide a structural C++ routine using indices without comparing standard value primitives with raw structural indices.

Part 3: Rapid Check-In Processing (Binary Search)

Once your array is completely sorted from Part 2, the final sequence state becomes:

```
int tickets[] = {1101, 1102, 2301, 4455, 5420, 8832, 9981};
```

An attendee presents ticket ID **4455** at the express lane counter.

Your Objective: Map out the search step trace using **Binary Search**. Complete the tracking values below to showcase how the boundaries contract on every integer division phase.

Binary Search Trace Worksheet:

Step	Low Index	High Index	Mid Index Formula	Array[Mid]	Action / Comparison Result
01	0	6	(0 + 6) / 2 = 3	4455	
02					
03					

Submission Guidelines: Provide completely standalone, compilable C++ code sections. Make sure your function parameters use clear array decayed pointers or standard constants where applicable. Ensure no bounds overflow errors exist.